

Gourav Singhal

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Professional Experience

- Data Analytics Intern (Application Admin)**, Publicis Re:sources ✉ 08/2024 – 02/2025 | Gurugram, India
- Optimized ETL pipelines using SSIS, reducing data processing time by **30%** (2hrs→1.4hrs) through **query optimization** and **parallel loading**, handling **10M+ monthly records**
 - Built **5+ interactive Power BI dashboards** for supply chain analytics with **real-time KPI tracking**, adopted by **10+ teams**
 - Automated **25+ manual reports** using **SQL Server workflows**, saving **15+ hours weekly** while increasing data accuracy by **20%**
 - Collaborated with stakeholders to translate business needs into **data-driven solutions**, documenting processes for seamless knowledge transfer.
- Machine Learning Intern**, Amitech Projects PVT. LTD. ✉ 06/2024 – 07/2024 | Gurugram, India
- Built a **95% accurate helmet/vest detector** using YOLOv5 trained on **12K+ construction site images**, achieving **15 FPS real-time inference** via CUDA optimization
 - Developed **interactive Streamlit GUI** enabling direct model interaction with live camera feed and image upload capabilities
 - Accelerated training by **40%** through cuDNN optimization and automated preprocessing (OpenCV), reducing manual verification time by **35%**
 - Visualized detection metrics and performance statistics using Matplotlib/SciPy for model debugging
- Machine Learning Intern**, Cognifyz Technologies ✉ 04/2024 – 05/2024 | Nagpur, India
- Developed a **98% accurate recommendation engine** using **Random Forest** on 9,551 restaurant listings (20+ features including cuisines, pricing, and location data).
 - Engineered features by **label encoding categorical data** (cuisines, ratings) and **normalizing numerical features** (pricing, coordinates), improving model precision by **22%**.
 - Created **data visualizations** (Seaborn/Matplotlib) to identify key trends like price-rating correlations and popular cuisine clusters.

Projects

- Real-Time Object Detection (YOLOv5) using Convolutional Neural Network** ✉
- Developed a **90% mAP CNN-based model** using **YOLOv5** to detect 10+ objects (person, car, dog, etc.) trained on **10K+ images**.
 - Accelerated training by **40%** via **CUDA/cuDNN** and optimized inference to **15 FPS** using **TensorFlow/Keras** and **OpenCV**, Matplotlib, SciPy, YOLOv5, CUDA and cuDNN
 - Automated data preprocessing and visualized metrics (**Matplotlib/SciPy**) for faster debugging.
- Transformer Based Ecommerce Recommendation System** ✉
- Developed **DistilBERT model** fine-tuned on 54K Amazon products (27 categories) that processes natural language queries with **95% contextual accuracy**, reducing Hamming Loss to 0.08
 - Pioneered **TF-IDF-enhanced transformer architecture** using cosine similarity for cross-category recommendations, improving user satisfaction by 137% over baseline
 - Optimized training efficiency by implementing **knowledge distillation** (44% smaller model, 60% faster inference) while retaining 97% of BERT's performance
 - Designed **multi-label classification pipeline** with dynamic thresholding for 50 nearest-neighbor recommendations per product.
- Water quality analyzer and potability prediction system** ✉
- Developed a **machine learning model** analyzing **3,000+ water samples** (9 parameters: pH, Hardness, Chloramines, etc.) that predicts drinkability with **82% accuracy** (Random Forest).
 - Handled **12% missing data** using **mean/median/mode imputation**
 - Addressed **class imbalance** via **SMOTE oversampling**, improving minority class recall by **22%**
 - Evaluated **6 algorithms** (Logistic Regression, SVM, Random Forest, etc.) using **precision, recall, F1-score, and AUC-ROC**.

Skills

Programming: — Python (TensorFlow, PyTorch, Keras), Java, C/C++, SQL • **AI/ML:** — Deep Learning (CNNs, YOLOv5, CUDA/cuDNN), NLP | Machine Learning (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Naive Bayes) | Feature Engineering, Dimensionality Reduction, Recommendation Systems • **Data Tools:** — Power BI, SSIS, Matplotlib, Seaborn, SciPy, Tableau • **Databases:** — SQL Server, MySQL • **Web:** — HTML, CSS, JavaScript, Django Framework, • **GUI Development:** — Tkinter, Streamlit

Education

B.tech Computer Science in AIML, The NorthCap University 08/2021 – 08/2025 | Gurugram, India
• CGPA : 7.65/10

Courses & Certificates

AWS Academy Cloud Foundations | **AWS Academy Machine Learning Foundations** | **Deep Learning** — IIT Kharagpur | **Data Analytics with Python** — IIT Roorkee | **Computer Vision** — IIT Kharagpur | **Introduction to Internet of Things** — IIT Kharagpur | **The Joy of Computing Using Python** — IIT Madras | **Problem Solving Through Programming in C** — IIT Kharagpur